# Vectaetos 1.1 – Formal Constraint Extension

Status: OFFICIAL

Ontology: Continuous (1.x Lineage)

Scope: Formal Constraints Only

APPENDIX A – Mathematical Formalism of Vectaetos 1.x

A1. Invariant Singularities
$\Sigma = \{\Sigma_1, \Sigma_2, \Sigma_3, \Sigma_4, \Sigma_5, \Sigma_6, \Sigma_7, \Sigma_8\}$
Each $\Sigma_i$ is ontologically invariant and non-hierarchical.

A2. Relational Structure
$R \in \mathbb{R}^{8\times 8}$
$R_{ij} = -R_{ji}$
$R_{ii} = 0$
R is antisymmetric and trace(R) = 0.

A3. Epistemic Field
$\Phi = (\Sigma, R)$
$\Phi$ is a relational topological state.

A4. Coherence Predicate
$K : \Phi \to \{0,1\}$
$K(\Phi) = 1 \Leftrightarrow H(\Phi) \geq \kappa$
K is a validation predicate only.

A5. Coherence Measure
$H(\Phi) = 1 - ( \Sigma\_\{i<j\} |R_{ij}| ) / Z$
$0 \leq H(\Phi) \leq 1$

A6. Quantum Epistemic Discontinuity (QE)
QE occurs if relational fragmentation appears in graph $G(\Phi)$.

A7. Vortex Operator
$V : \Phi \to \Phi'$
Preserves antisymmetry.
No optimization or global objective allowed.
No guaranteed attractor.

A8. Zero Agency Constraint
No operator $O : \Phi \to \Phi$ such that
$O(\Phi) = \text{argmin } F(\Phi)$ or $\text{argmax } F(\Phi)$
Reward mechanisms prohibited.

A9. Acyclic System Constraint
System layer graph must be a DAG.

No Projection → Φ feedback.
No Memory → Vortex influence.

A10. Stateless Execution
Independent executions must not influence each other.

A11. Silence Legitimacy
$\exists\, \Phi$ such that $P(\Phi) = \varnothing$

A12. Memory Non-Influence
$\Phi' \neq f(\Phi, M)$

Formal Definition:
Vectaetos 1.x = $(\Sigma, R, K, \kappa, V, P)$
Subject to antisymmetry, non-optimization, DAG architecture,
stateless execution, memory non-influence, silence legitimacy.